CourseWork 2025

Design and validation of a grid connected three phase full bridge inverter

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1. Definition of the system and system specifications.

The system under study is a **three-phase, grid-connected full-bridge inverter** designed to interface a DC energy source (in our case a photovoltaic array or battery energy storage systems) with the AC grid, with the objective of **exchanging power** (this can be absorbing or injecting power).

The inverter **converts a constant DC voltage into controlled AC currents** that are injected onto the grid with sinusoidal waveforms synchronous with the power grid.

This system includes:

- A **constant DC voltage source**. (In our case split by two VDC/2 sources to be able to have a neutral reference point at 0 and be able to generate the desired AC signal).
- A **three-phase full bridge inverter** (composed of three independent half -bridges).
- **PWM modulation** at each phase.
- **Filtering inductors (L),** between the inverter and the grid**.**
- An **AC grid** modeled “ideally” as a three-phase voltage source.
- A current control system.
-

It will be designed according to the following **parameters**:

-DC link voltage (VDC) = 800V

- AC grid voltage (VAC) = 230V

- Switching frequency (fs) = 10kHz.

- Maximum current ripple ΔI = 2A

- Grid frequency = 50 Hz (standard EU)

- Rated Power = 8kW.

- DC voltage ripple ΔV = 0.5V

- RL = 0.036 ohm.

<u>Part 1</u>: Inverter with a constant DC source and AC current control.

2.1- Inductor L design.

To be able to filter out the high frequency switching harmonics from the desired L.F components that are responsible for the power transfer we must **implement an L filter** at the output of our inverter. The correct design of this inductance is important as a very small "L" will allow an easier power transfer, but it will give a poor filtering, and a very large "L" will allow better H.F. harmonics filtering but it will limit power transfer. This relationship can be observed in the following equation:

$$P_{AC} - jQ_{AC} \approx \frac{V_{AC}V_{INV}\gamma}{\omega L} - j\frac{V_{AC}V_{INV} - V_{AC}^{2}}{\omega L}$$

Therefore, we will try to design the **smallest L possible that still fulfills the filtering requirements.**

To do so we assume the worst-case scenario and we design for the maximum possible ripple:

$\Delta I = \frac{V_{DC}}{2}\frac{1}{L}\frac{(T_S/2)}{2}$ and from this expression we get: $L = \frac{T_S V_{DC}}{8\Delta I}$

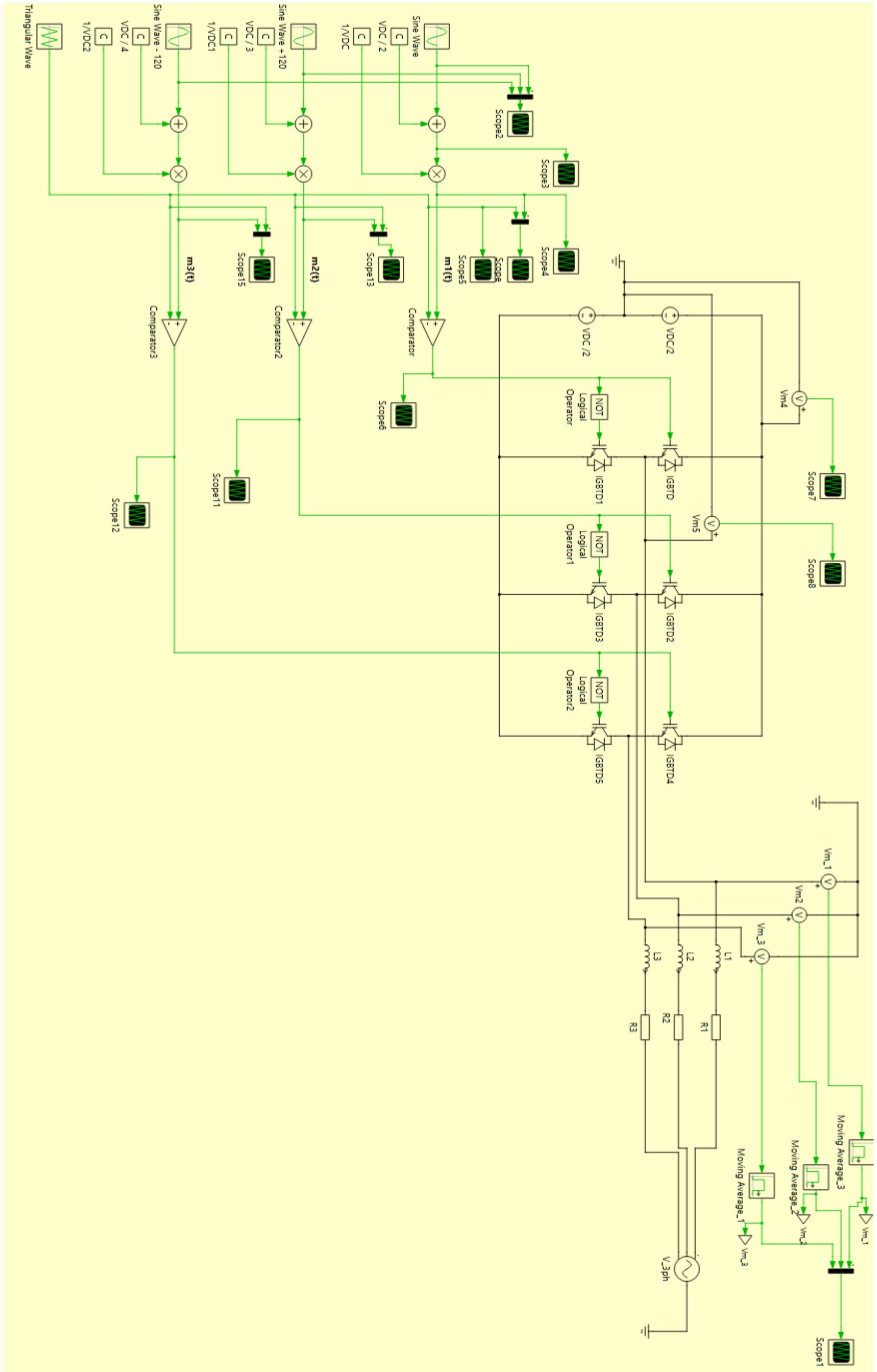
Therefore:

$$\boldsymbol{L = \frac{\left(\frac{1}{fs}\right) * VDC}{8 * \Delta I/2} = \frac{0.0001 * 800}{8} = 0.01\, H = 10mH}$$

This is convenient since **10mH is a standard value for inductances**, and it will be readily available for implementation.

*Please note this formula is slightly different from the theory since we consider peak to peak current and we use a full Ts instead of half for the calculation.

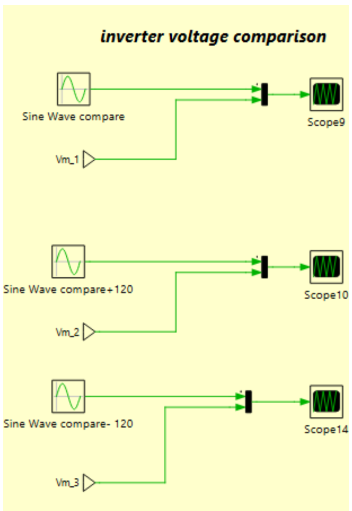
2.2- PLECS model of the three-phase full bridge inverter with independent PWM modulation.

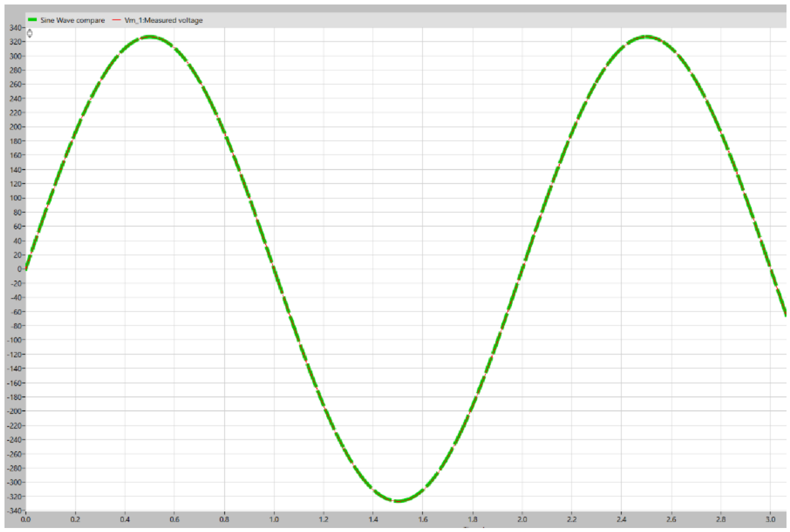


2.3- Validation of results and choice of L.

To validate the design and check that the choice of L is correct, we simulate the following waveforms:

-Comparison of **resulting voltages and reference voltages given to the PWM modulator**:





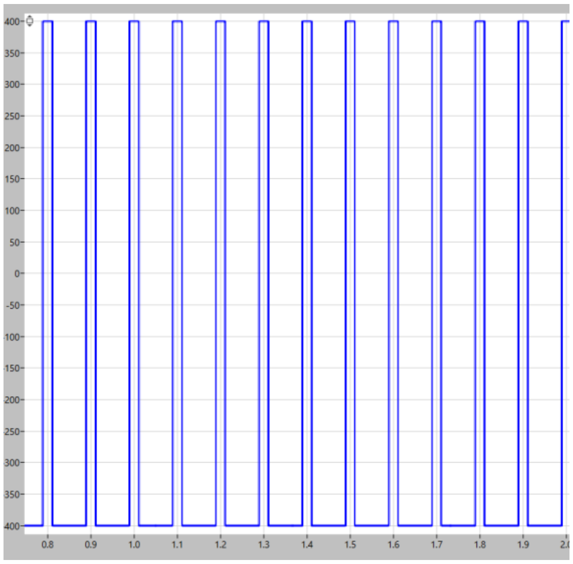
In this graph I have shown the overlapping of the signal of the reference voltage waveform we give the PWM modulator (green line), that knowing that the AC RMS voltage is 230V, we know it will have a peak at 326.6V and a frequency of 50Hz, as is standard in EU grids, with the voltage signal resulting after the three-phase inverter.

This graph is only for one of the phases. For the signals with the phase shift of 2 π / 3 and 4 π / 3 the results are equivalent, and phase shifted as are the reference signals.

For this reason, it is concluded that the **three-phase full bridge inverter generated the desired AC voltage to match the AC grid.**

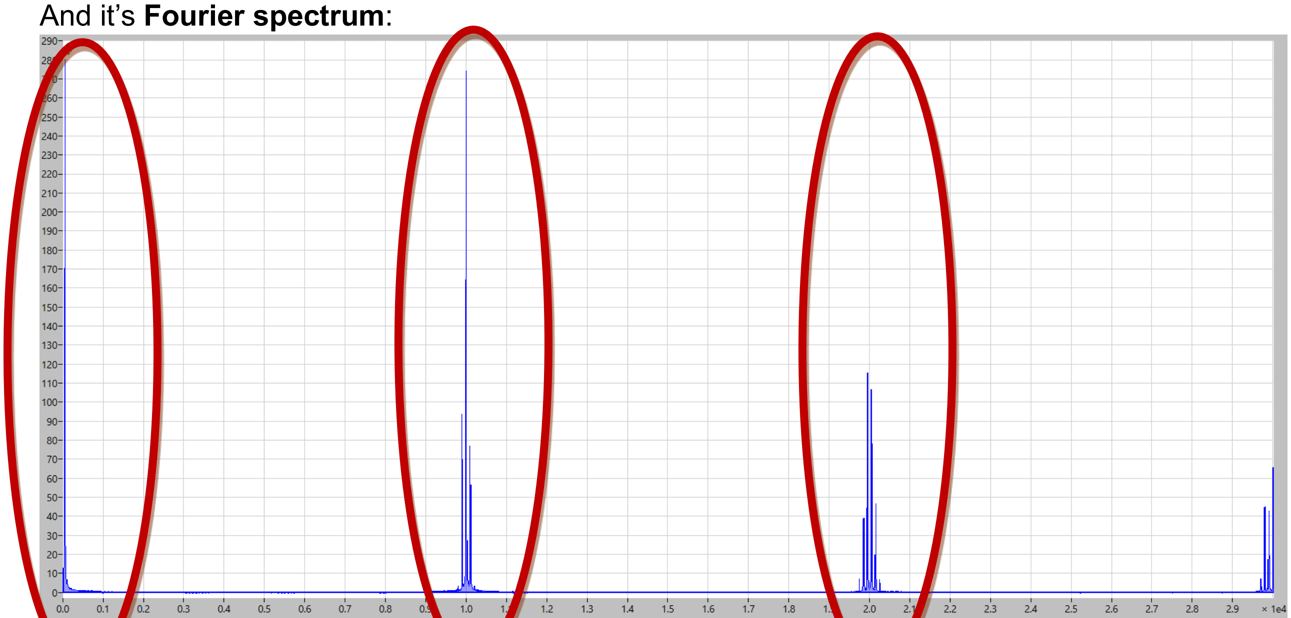
-Waveforms of the generated unfiltered phase voltages:

For reference we take the generated voltage signal for the first phase.



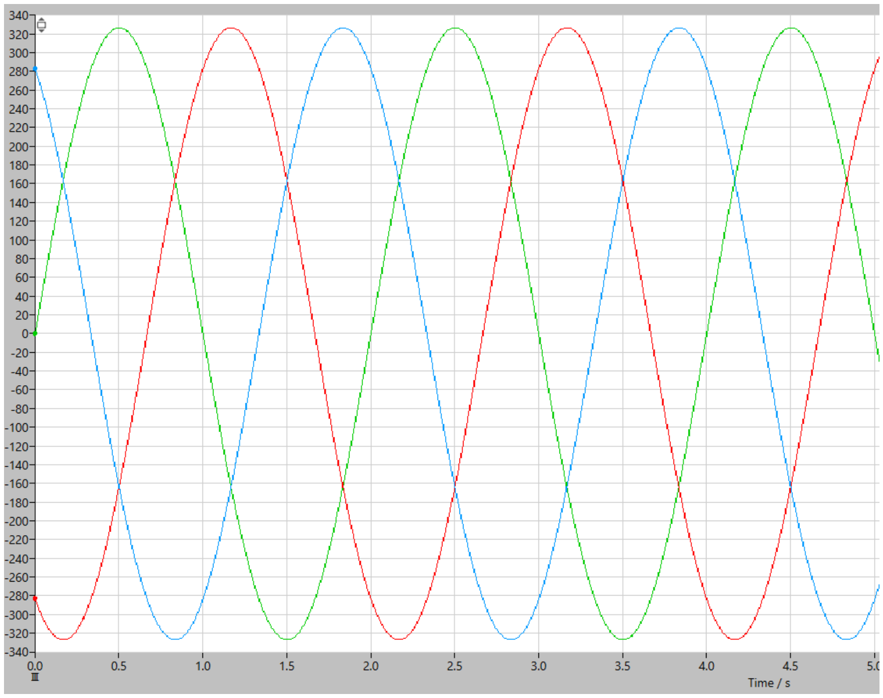
- In this case we see a **waveform switching between VDC/2 and -VDC/2**, the amount of time it spends at the higher point and the time it spends at the lower time will determine **the average voltage over a time period**, in this way, after filtering, we obtain a smooth sinusoidal waveform equal to the reference signal given to the PWM modulation.

And it's **Fourier spectrum**:



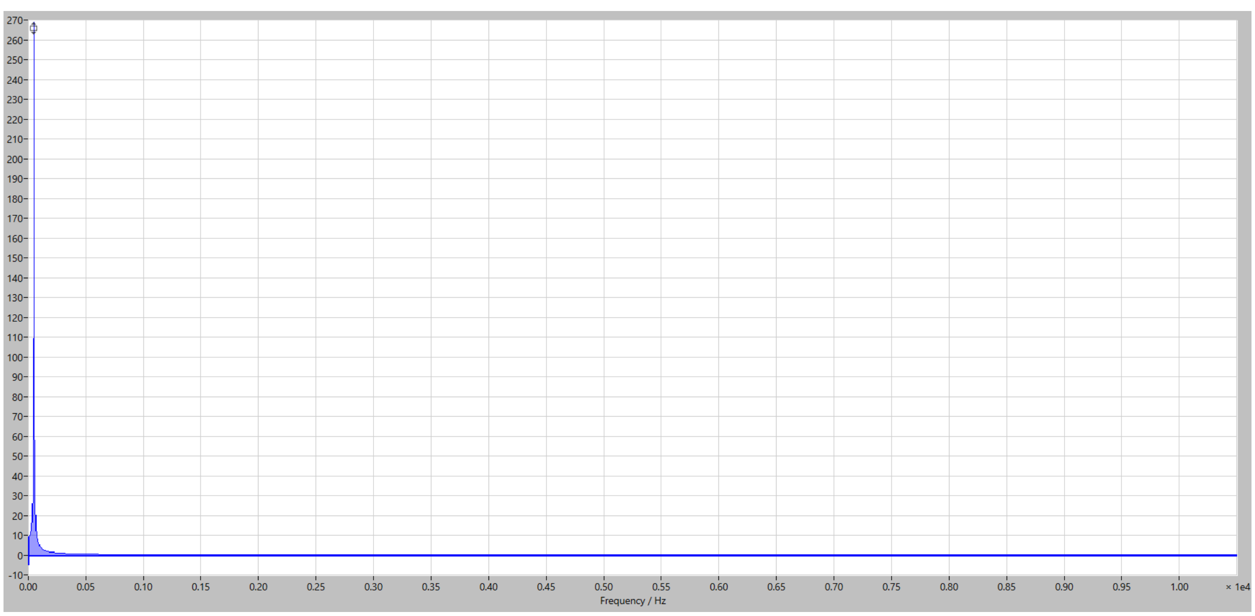
In this case we see the fundamental harmonics at 50 Hz, these are the L.F components that contribute to power transfer. At 10kHz, which is the switching frequency, we observe there is also a H.F component which will be necessary **to filter out to obtain the desired performance in the three-phase inverter.** We also expect to see switching harmonics at higher frequencies, multiples of the fundamental switching frequency (fs*2, fs*3, etc.).

- Waveform of the three-phase voltage (Line to N), after filtering:



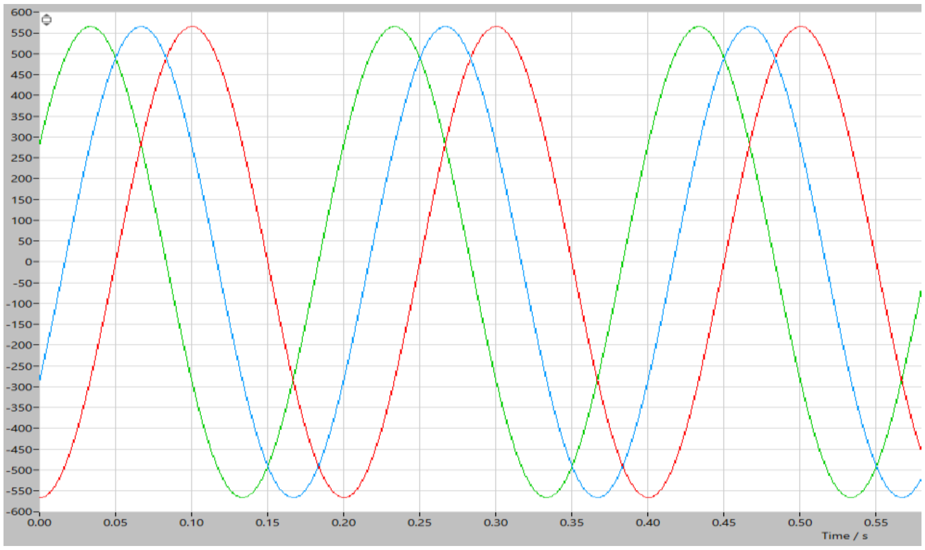
We observe the three waveforms that are **identical, but phase shifted by 2 π/3 between them.** As mentioned before, after filtering, we obtain these smooth sinusoidal waveforms with amplitude 326.6V and 50Hz frequency, that are equal to the reference voltages given to the PWM, and those coming from the ideal AC grid we have modeled in PLECS.

- Fourier spectrum:



Here we observe the L.F component is present at 50Hz but the **H.F switching harmonics we found at 10kHz have been filtered out by the inductance L.**

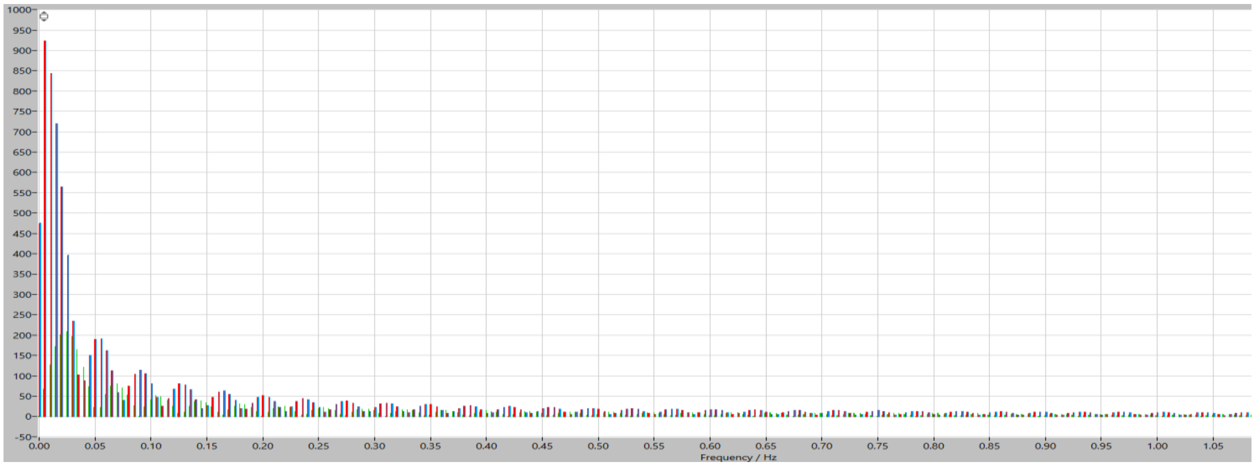
-Waveform of the three-phase voltage (Line to Line), after filtering:



This graph represents the difference in electric potential between two lines at the output of the inverter, as expected from the theory, we get a line-to-line voltage oscillating between a peak equal to 565.6V and a minimum of -565.6V, which is the peak value when VRMS_line-to-line is 400V which is standard in the EU.

In this case we take line to line Vab, but we expect the same result for Vbc and Vac with the corresponding phase shifts.

- Fourier spectrum:

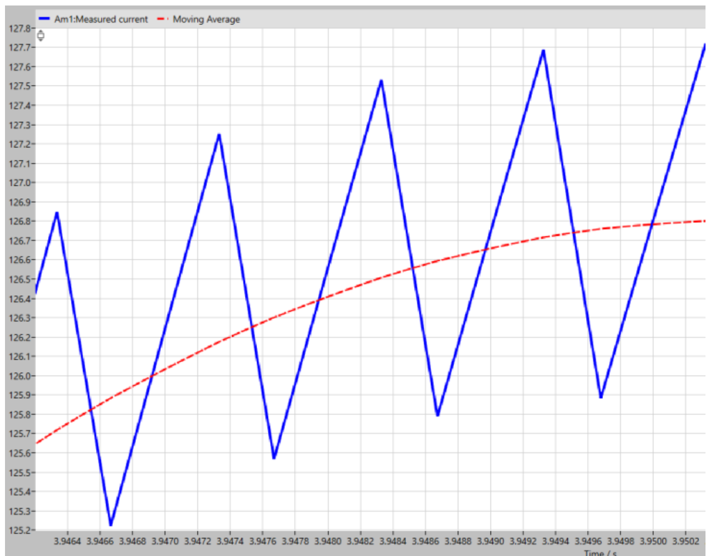


In this case we see a higher number of harmonics around 50Hz, compared to the V line to N case the value it reaches is around 3 times the fundamental harmonics voltage (around 300V). As in the previous case, the high frequency harmonics from the inverter switching are filtered out by L.

-Choice of L:

As previously seen, with an L= 10mH we can fulfill the filtering requirements to eliminate the H.F components of the voltage signal coming from the three-phase, full-bridge, inverter controlled via PWM modulation.

A second condition must be validated, that regarding the maximum current ripple. To do so we measure the current waveform, and we calculate the peak-to-peak ripple using the scope that shows the current waveform and the cursors:



Data

Name		Cursor 1	Cursor 2	Delta
Time		0.394932	0.394968	3.54882e-05
Plot 1				
Am1:Measured current		127.688	125.891	-1.79639
Moving Average		126.717	126.761	0.0443215

With the selected L, we obtain a current ripple at around 1.80A measured with our cursors, this way we confirm that it is within our acceptable range, **so the selection of L is acceptable**.

2.4- Transfer function calculation

To be able to calculate a transfer function for the plant we must consider each of the voltage phases independently, later we will consider the coupling effects between the phases when designing the controller.

The dynamics of our system can be modeled as:

$$L * \left(\frac{dIx(t)}{dt}\right) + R * Ix(t) = Vinv(t) - VACgridx(t)$$

Where x = a, b or c for each corresponding line.

Then in the Laplace domain:

$$(sL + R) * Ix(s) = Vinv(s) - VACgrid(s)$$

$$G(s) = \left(\frac{Ix(s)}{Vpwm(s)}\right) = \frac{1}{R + sL} = \frac{1/L}{L * \left(s + \frac{R}{L}\right)}$$

Where **G(s) is the plant transfer function** having as input the Vpwm(t) (voltage demand to the PWM) and the corresponding AC grid current in phase (Ix(s)) as output.

2.5- Controller design:

We would like to implement a controller to have **zero steady-state error**, a **damping factor of 0.8 and a natural frequency f0=500Hz**. We will assume to use a PI controller to have the proportional and integral properties of the controller:

$$C1(s) = Kp + \frac{Ki}{s} = Kp * \frac{s + a}{s} \ where \ a = \frac{Ki}{Kp}$$

And from the plant transfer function we simplify as:

$$G(s) = \frac{B}{s + D} \ where \ B = \frac{1}{L} \ and \ D = \frac{R}{L}$$

And therefore, the open loop T.F will be:

$$G1(s) = Kp * \left(\frac{s + a}{s}\right) * \left(\frac{B}{s + D}\right)$$

$$G1(s) = (Kp * B * \frac{(s + a)}{s * (s + D)})$$

And for the closed loop function:

$$G1cl(s) = \frac{G1(s)}{1 + G1(s)} = \frac{Kp * B * (s + a)}{s * (s + D) + Kp * B * (s + a)} = \frac{Kp * B * (s + a)}{s^2 + s * (Kp * B + D) + Kp * B * a}$$

Comparing to the standard order characteristic equation:

$$s^2 + 2\xi\omega o * s + \omega o^2 = 0$$

$$Kp * B * a = \omega 0^2$$

$$Kp * B + D = 2\xi\omega o$$

Having **ωo=2*π*fo=3141.59 rad/s and ξ = 0.8:**

$$Kp = 50.22$$

$$a = 1964.90 \sim 1965 \text{ and then } Ki = a * Kp = 98.697$$

And therefore, the final PI controller will be:

$$Gc1(s) = 50.22 * \frac{s + 1965}{s}$$

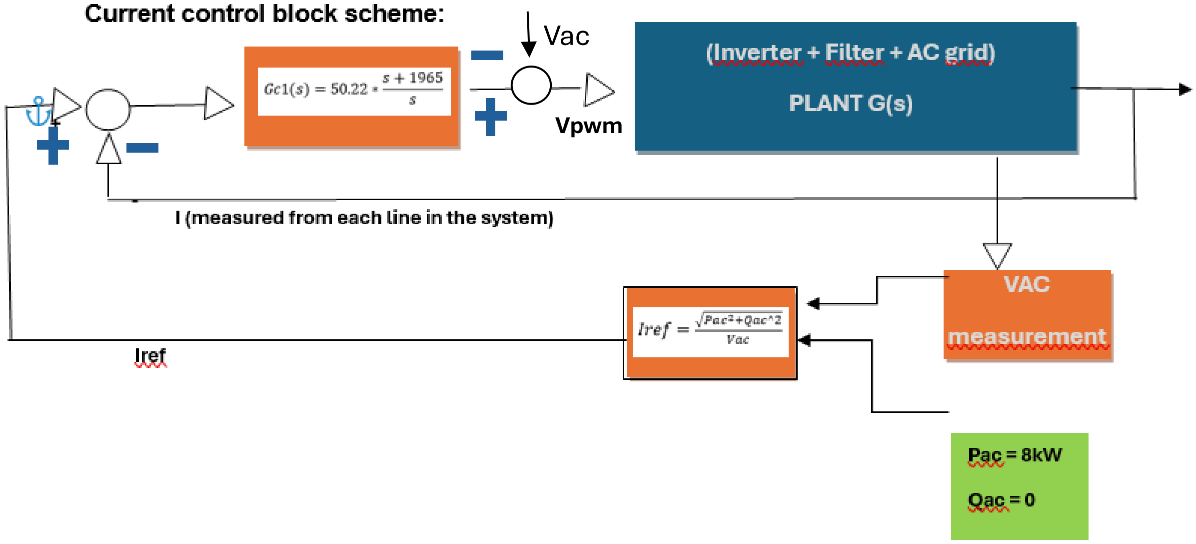
Also, it is important to calculate the current references we feed into the controller considering the desired active and reactive power. We want to exchange 8kW of power according to the problems specifications. Let's assume we only want to exchange active power with the grid; this is the most usual case.

Pac = 8kW then $Iref = \frac{Pac\sqrt{2}}{3Vac}$ $where\ Vac\_rms = 230V\ in\ EU\ then\ Iref = 16.35\ A$

Qac = 0;

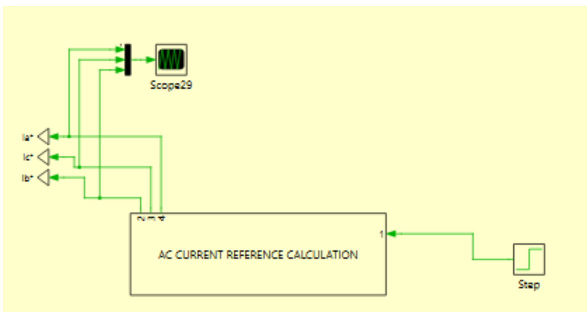
Then we must set **Iref=16.35A** as a peak for the current reference for each of the PI controllers. These will be shifted by 120 degrees from each other.

Current control block scheme:

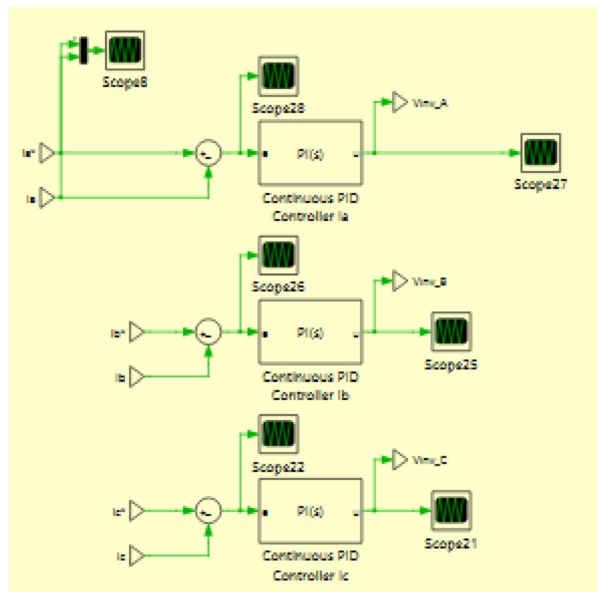


2.6- Implementation of the controller in PLECS.

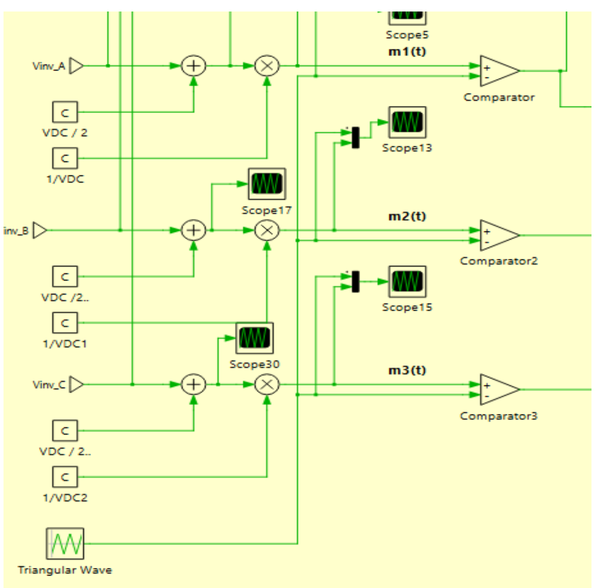
To be able to **implement the PI controller** we have designed in PLECS we have to create three **current references (Ia*,Ib*,Ic*)** to be able to feed them back into the control loop, by comparing it to the measured (Ia,Ib,Ic) and creating an error signal to input into the PI controller we have previously designed using our plant model. Our PI controller will then create a Voltage signal we can use as input for our PWM modulation.



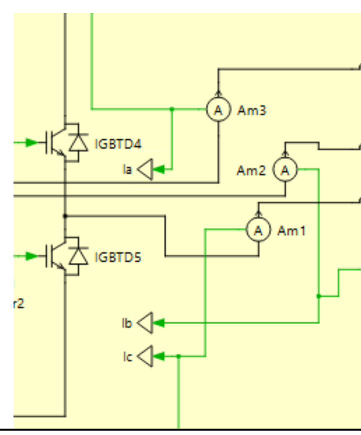
This subsystem creates the (Ia*,Ib*,Ic*) from the Pac input. This block uses the formula we used to calculate them analytically considering null reactive power and Pac= 8kW.



Implementation of the PID controllers, output is Vinv. Also, a saturation of +VDC and -VDC has been added and anti-windup by Back-calculation

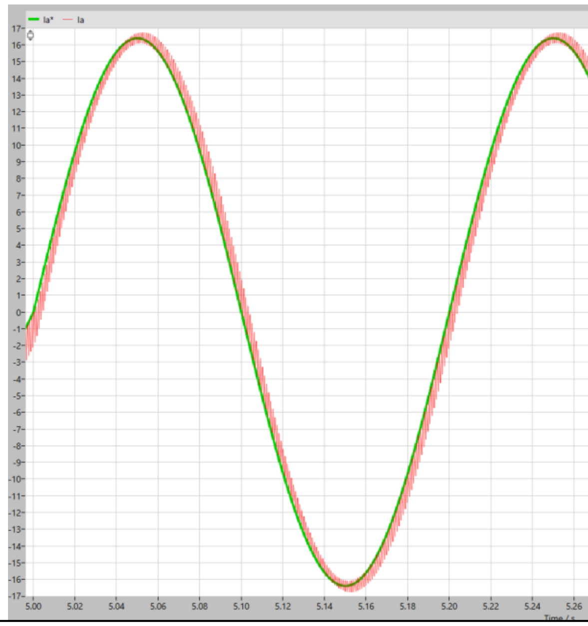


We then use Vinv (for each phase a,b,c) as reference signals for P.W.M modulation.



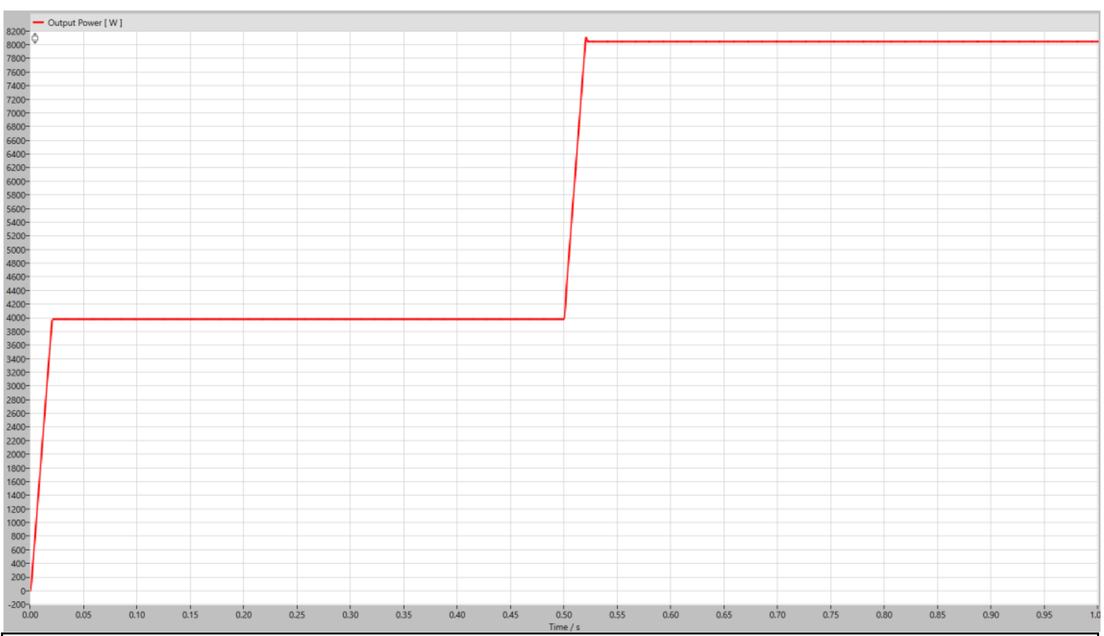
Measurement of real currents (Ia,Ib,Ic) to compute e(t) for the PI control

As results from our control in close loop on the grid-connected three phase inverter, we observe:
The real current in each of the lines (Ia, Ib, Ic) follows closely the reference currents (Ia*, Ib*, Ic*):



We observe a **close following** of the real current (with its visible current ripple always under 2A) to the Iref we have calculated. This means the close loop system converges. This graph plots only Ia* and Ia but we observe the same behavior with the comparison for the other line currents (b and c).

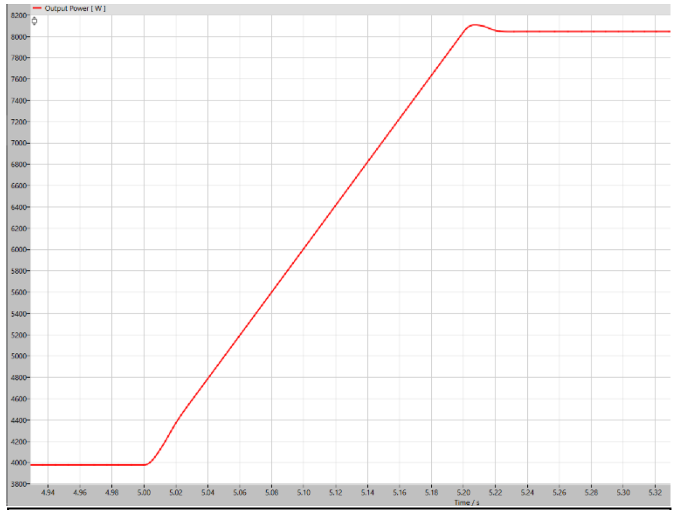
Controller power output during transients 0 to P/2 and P/2 to P = 8kW.



The average output power curve reaches the desired values in steady state (4000kW and 8000kW) with **minimal steady state error**. It reaches **3980W** for the first power reference value and **8043W** for the second one. This error is around 0.5%, which is considered acceptable.

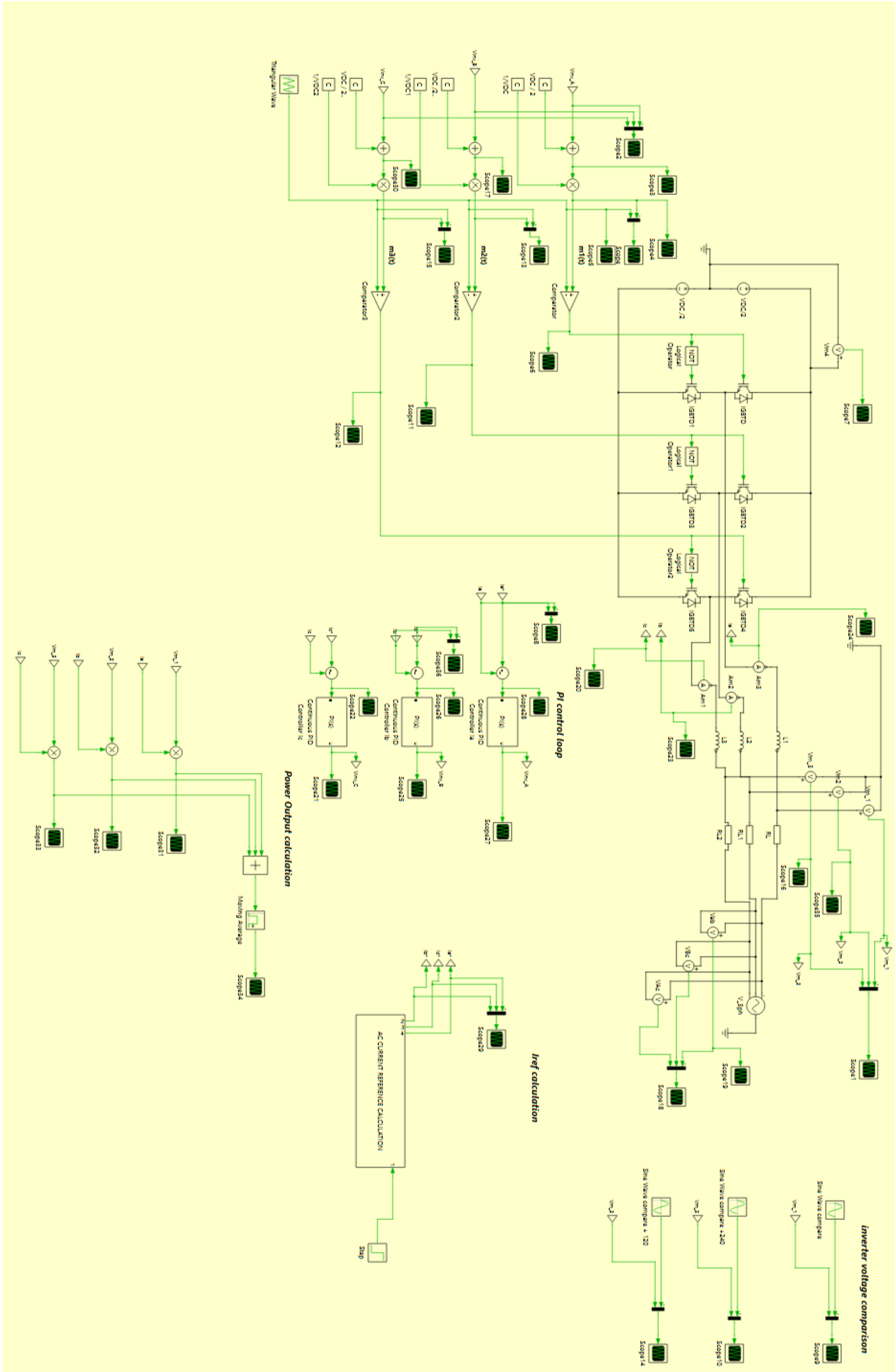
Behaviour during transients:

During transient from 0 to P/2 we can observe a quick rise time of around 0.02 seconds and a very slight overshoot and oscillation.



During transient from P/2 to P we can observe a quick rise time of around 0.02 seconds and a larger overshoot and oscillation. This **overshoot value reaches around 8110 W, which is around a 1.5% over target value,** this is considered a good compromise between response speed and damping.

We consider that the close loop control system is operating correctly since we are controlling the output and it reaches the desired values with minimal steady state error quickly and it also dampens considerably the oscillations after the transients. If we proposed a controller with a lower dampening factor, we would expect to see larger overshoots and oscillations.


PI control loop
Power Output calculation
Iref calculation
inverter voltage comparison

3. <u>Part 2:</u> Inverter with DC link capacitor, DC voltage control and AC current control.

3.1- DC link capacitor design.

For this second scheme we want to implement a DC link capacitor in parallel with the inverter. To do so, we must first design capacitor C with the correct values, these will depend on the current flowing through the capacitor and the maximum allowable voltage ripple (ΔV=0.5).

The voltage ripple will be represented as:

$$\Delta V = \frac{\Delta Q}{C} \;\; where\; \Delta Q = \frac{Ts * \Delta ic}{4}$$

Then using this approximation of Ic as a square wave when we have a high switching frequency:

$$C = \frac{Idc}{4 * fs * \Delta V} = \frac{PAC}{4 * fs * Vdc * \Delta V} \; by\; using\; Ohm's\; law.$$

Therefore, in our case:

$$C = \frac{8000}{4 * 10000 * 800 * 0.5/2} = 1mF$$

We will need a capacitance of around **1 mF.** In this case, if we consider the ripple voltage not a critical requirement we would choose the **nearest standard value for the capacitance, which is 1mF itself**, this would be practical as they are readily available. In the other hand, if the voltage ripple is critical, we would choose the **nearest larger standard capacitance, which is 1.2 mF.** During this report **we will assume 1mF as it matches the calculations**.

3.2- Transfer function calculation

We need to calculate the transfer function of the plant represented by the converter, to do this we consider the **input the Vdc voltage calculated and the Pac as output**. This transfer function will be necessary when designing the voltage control loop of the inverter. We will keep the voltage control loop as the outside (slower) loop and an AC current control loop as the inner (faster) loop, this way we neglect the current control loop during the design of the voltage control loop. The output of the voltage control (Pac) will be necessary to implement the AC current control loop.

Since we know, from power balance principle, that: $Pc = Ppv - Pac$

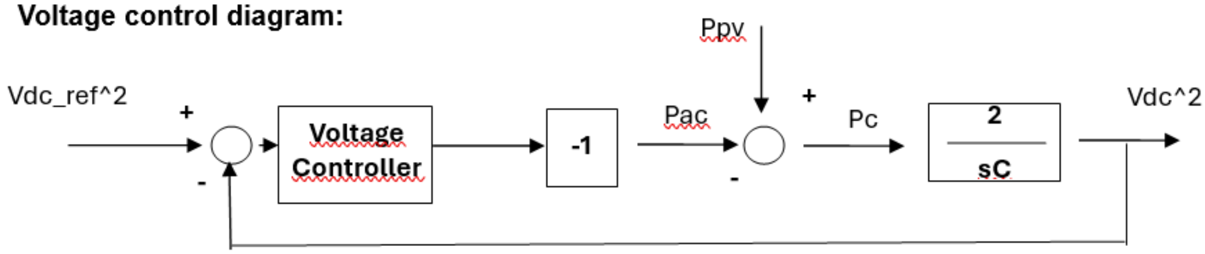
And we know the energy stored in the capacitor (Ec): $Ec = \frac{1}{2} * C * Vdc^2$

Therefore: $Vdc^2 = 2 * \frac{Pc}{sC}$ *in the Laplace domain* $\frac{Vdc^2}{Pc} = G(s) = \frac{2}{sC}$

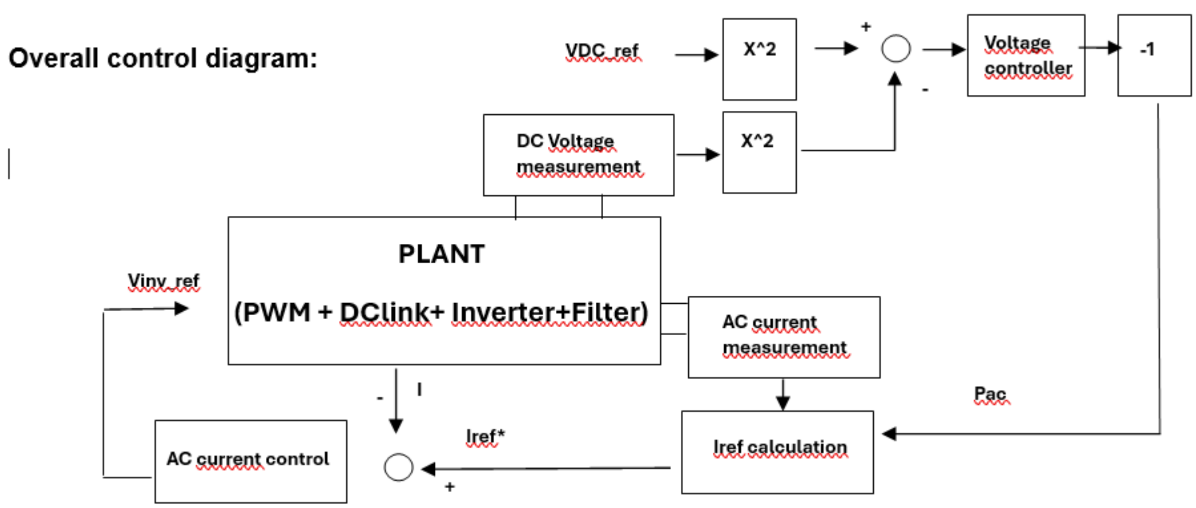
and in this case:

$$G(s) = \frac{2}{s * 1000 * 10^{-6}}$$

Voltage control diagram:



Overall control diagram:



3.3- Voltage controller design

Having calculated our plant model G(s) and assuming we implement a PI controller the open loop transfer function will be:

$$G1(s) = C2(s) * G(s) = Kp * \frac{s+a}{s} * \frac{2}{sC} \;\; where\ a = Ki/Kp$$

And therefore, our closed loop T.F:

$$G1cl(s) = \frac{G1(s)}{1+G1(s)} = 2Kp * \frac{s+a}{s^2 * C + 2 * Kp * s + 2 * Kp * a}$$

And from the characteristic polynomial equivalence:

$$Kp = \xi * wn * C = 0.85 * 2 * Pi * 10 * 1000 * 10^{-6} = 0.0537$$

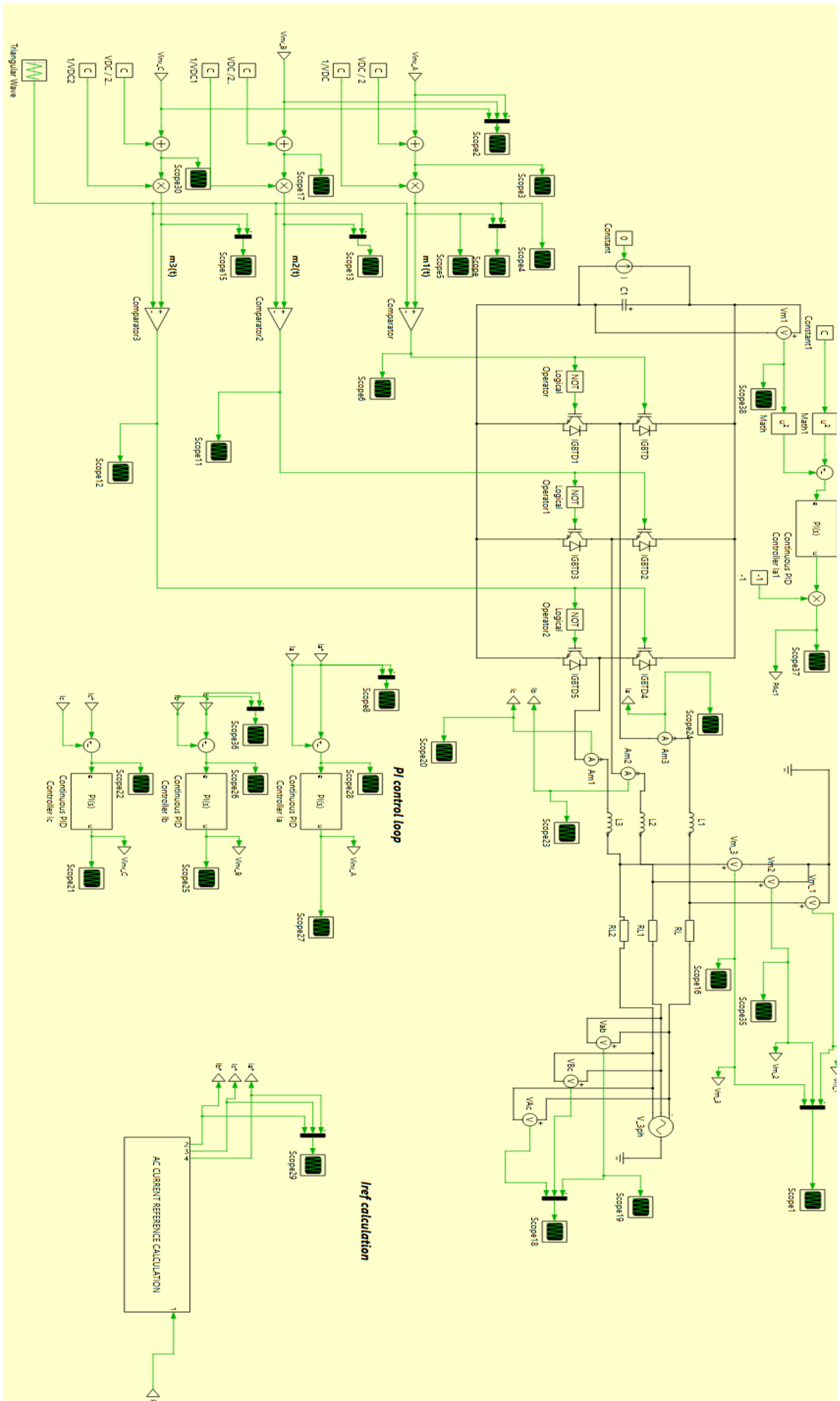
$$a = \frac{wn}{2 * \xi} = 2 * Pi * \frac{10}{2 * 0.85} = 36.96$$

Therefore, the resulting PI controller:

$$C2(s) = 0.0537 * \frac{s + 36.96}{s}$$

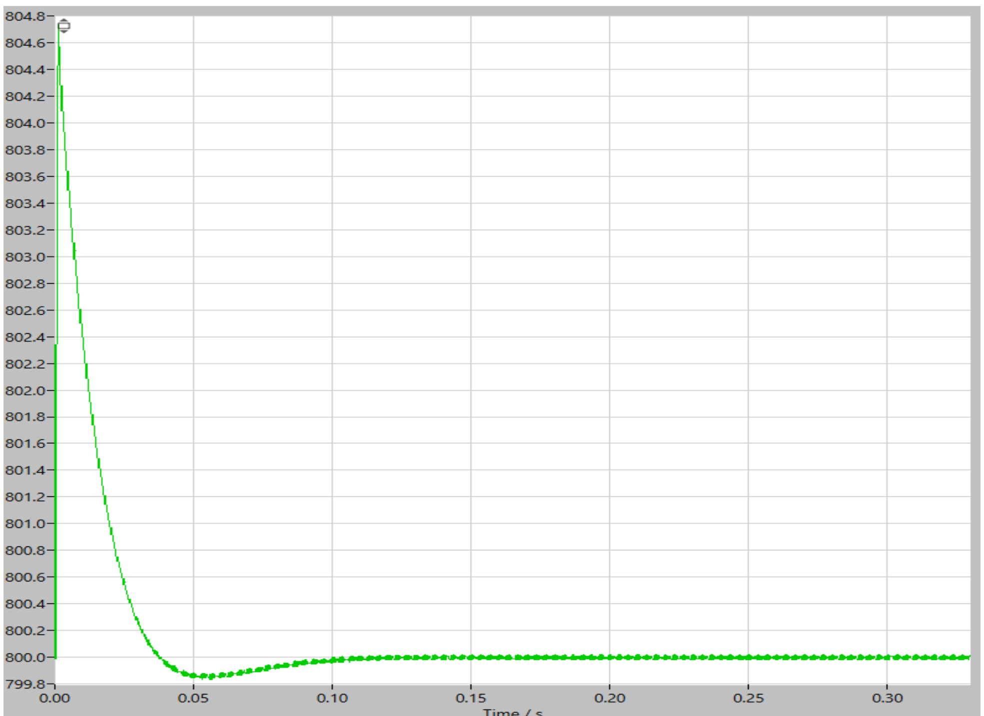
3.4- Overall control implementation

We implement both AC current and DC voltage control loops.



We set **Vdc_ref as 800V** for it to be large enough, we also had to set initial VC through the capacitor at a value Vdc, otherwise the control reacts aggressively, this initial VC gives us a more realistic start-up point, normally the capacitors are pre-charged to avoid the capacitor behaving like a short-circuit and having large inrush currents that could potentially damage the hardware. In this case we **leave the current in parallel with the capacitor, Idc = 0**.

And in closed loop we get Vdc across the capacitor:



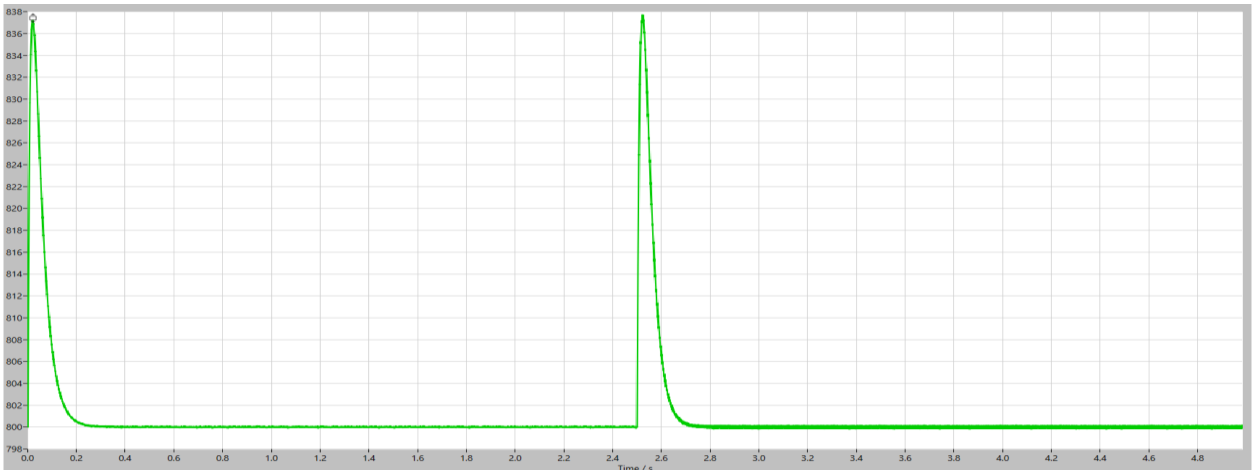
We can see that **Vdc reaches 800V** with minimal error, which was the Vdc reference voltage, in steady state, and it does so in around 0.1s. We can see there is a slight undershoot and oscillation but heavily dampened by the system. We conclude then the PI control is correctly designed.

3.5- Vdc convergence during transients:

$$0\ to \frac{Ir}{2}\ and \frac{Ir}{2}\ to\ Ir\ , where\ Ir\ is\ the\ rated\ current\ Ir = \frac{8000}{VDC} = 10A\ as\ \boldsymbol{we\ have\ chosen\ 800V}\ as\ reference$$

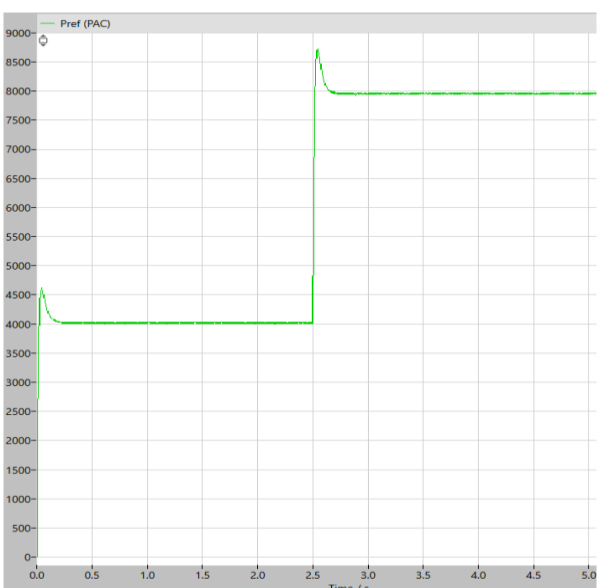
Applying this input to the current source:

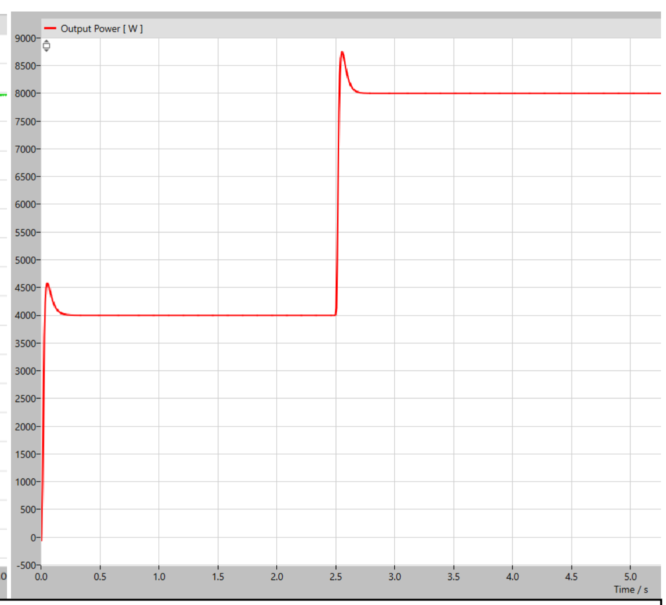
-Voltage curve (Vc):



We can see that at the transient periods (imposed at t=0s and t=2.5s) there are **two peaks reaching around 840V**, this happens because there is an **abrupt change in the current** and the capacitor cannot charge instantaneously, and so there is a peak that when the capacitor is fully charged again, it goes down to our imposed reference voltage (800V) thanks to our control loop. We can observe that in steady state the **voltage converges to our reference with minimal steady state error and minimal oscillations, so the PI controller is well designed.** If we wanted a faster response we could design the controller with a lower damping factor.

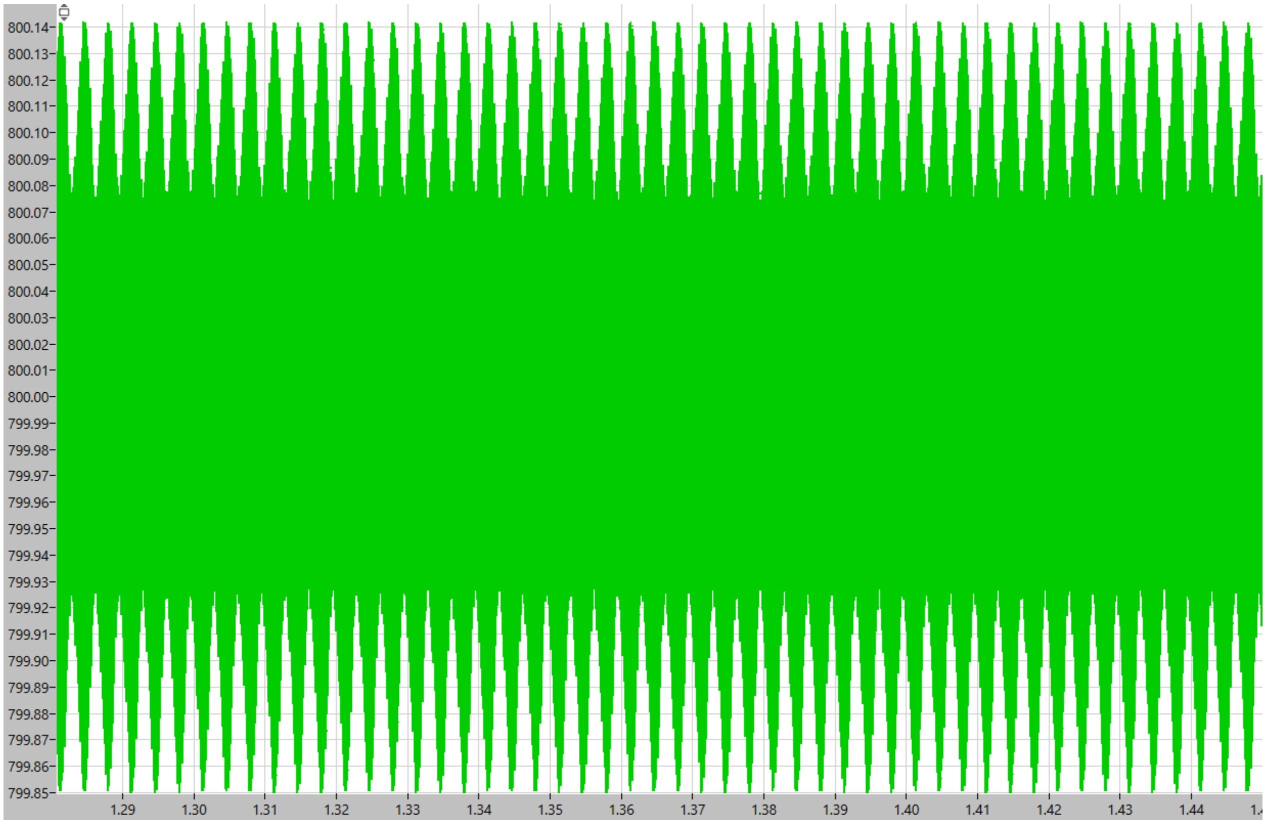
-Reference power (Pref)/ real Output power (Pout):





Here we can see the comparison between the reference power (PAC) that comes from the voltage controller and the real power output of the system, this way it is concluded that **there is a good tracking of the reference and that the system reaches the desired power.** The steady state error is so small it is negligible. The two peaks in power come from the transient behavior observed in Vdc and the oscillations are dampened reaching a good balance between response speed and setting time.

3.6- Vdc ripple when operating at full DC current (steady state):



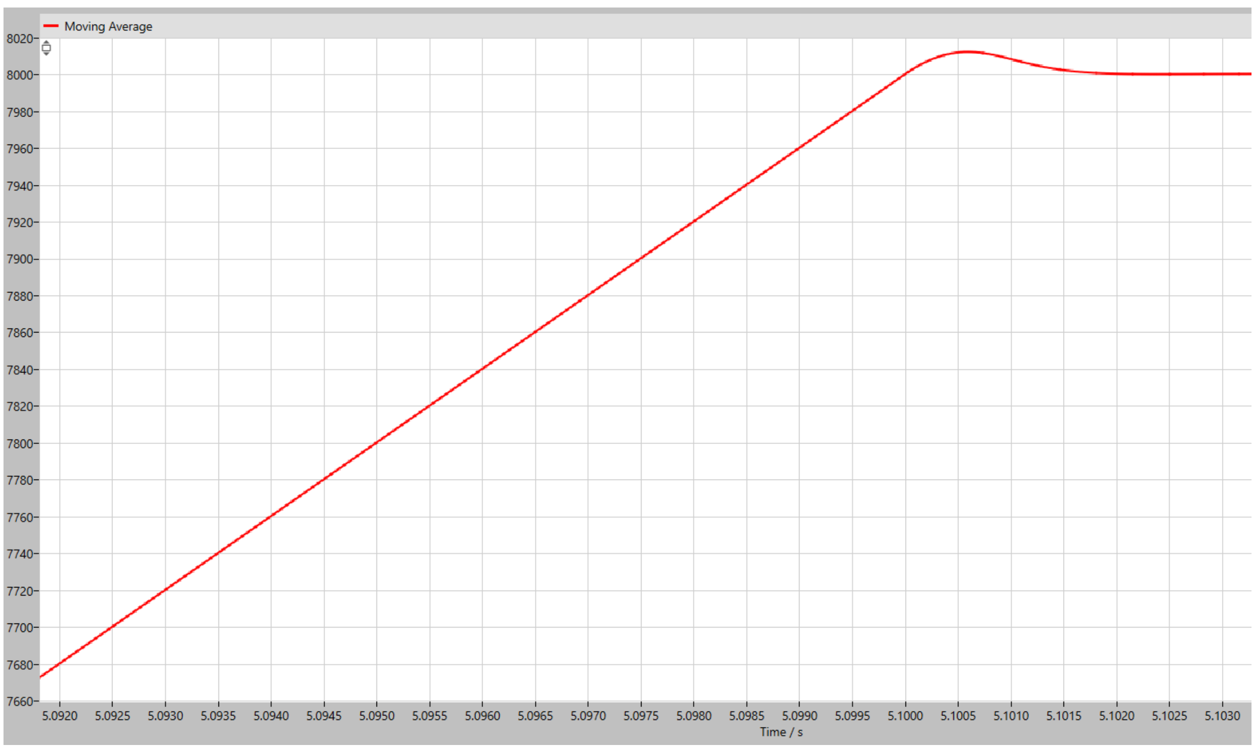
The voltage ripple from Vdc across the capacitor stays at around a maximum of 0.3V which is inside the required ΔV=0.5V. If we had more stringent requirements we could implement the system with a larger capacitance, and therefore the matching Kp and Ki coefficients to get a similar response in terms of speed and oscillations.

4. Conclusion

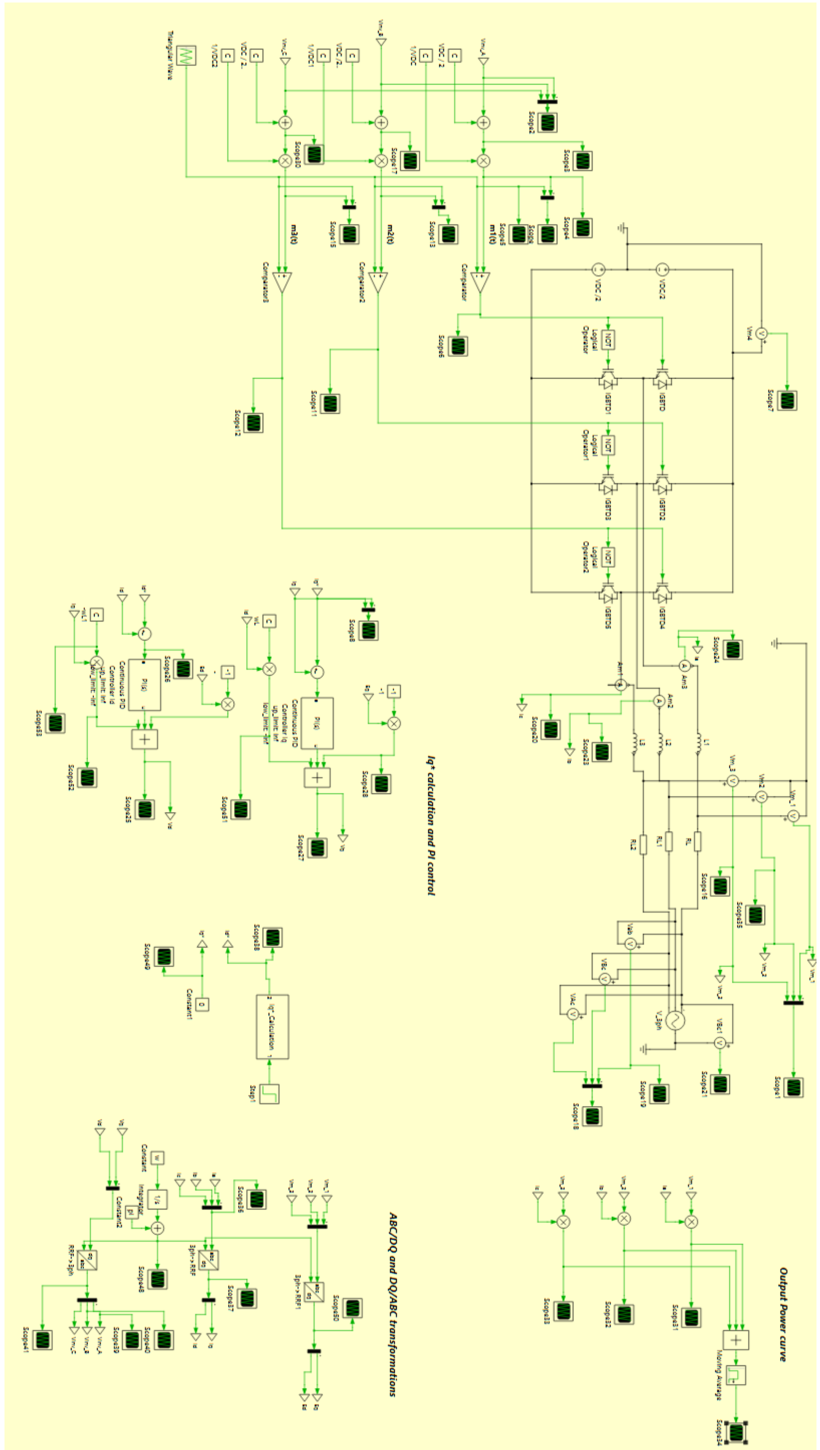
It is concluded that the **design of the components and control loops is successful** because after validation the results coincide with the expected or required behaviors.

In the case of the need to implement this system in a **real-world system** it would have to be considered that the **data we assumed as known from the AC grid would have to be measured and processed through PLL or other methodologies**. Also, there would be the need to use the obtained phase angle to perform the ABC to DQ conversion using **Park and Clark transforms** to be able to control the currents as a constant and perform a decoupled control of active and reactive power to generate the voltage references for PWM modulation. This would give improved performance and stability as well as it allows to work with an AC source (the grid) where we cannot assume to know the phase angle and exact behavior.

Additionally, this control scheme has also been tested to confirm the improvement in behavior of PART 1 of the project when applying DQ control to the AC current control loop.



With the same control parameters as before we observe that the most critical transient 4kW to 8kW exhibits better behavior (**less overshoot and less oscillation**) half the rise time, therefore we can say that for critical control applications it may be better to apply the control in the DQ domain. This is because by transforming an AC signal to a DC signal we leverage the improved performance of PI controller in a DC signal.



Bibliography:

Slides from "509674 - POWER ELECTRONICS" – Prof. Zanchetta Pericle

"Modern power electronics and AC drives"-Bimal K. Bose